Joe DiGiovanni

Data Scientist - AI Systems

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Summary

Data Scientist with an M.S. in Applied Data Science from UChicago. Focused on finding structure in ambiguous, high-variance problems and building systems to translate complex analytical outputs into practical insights. Experience spans causal inference research, LLM-powered product development, and end-to-end pipeline design.

Experience

Founder & Lead Data Scientist **Sep 2025 — Present**
FullCount AI *Remote*

- Built a public analytics platform querying over two-million rows of high-dimensional Statcast data, enabling non-technical users to create customized interactive visualizations with self-serve charting and filters across pitching and hitting metrics.
- Diagnosed a core reliability failure in LLM-driven data querying of hallucinated SQL results and re-architected the system around a deterministic DuckDB routing layer, establishing correctness as a hard constraint rather than a best-effort outcome.
- Conducted structured discovery with alpha testers to identify trust and consistency failures in existing AI analytics tools, replacing raw LLM output with modular UI components that keep results verifiable and repeatable for non-technical users.

Development Data Coordinator **Jul 2026 — Nov 2026**
Para Los Niños *Los Angeles, CA*

- Prepared and validated constituent giving data by querying, cleaning, and reconciling records in Raiser's Edge NXT, then produced analytical reports, data visualizations, and fundraising projections used by senior leadership across active campaigns.
- Operationalized RFM-based constituent segmentation through a third-party analytics platform, identifying high-value and lapsed donors, prioritizing outreach, and translating attrition patterns into retention strategies with development leadership.
- Strengthened fundraising data integrity by auditing constituent and gift records, resolving duplicates, correcting campaign and appeal attribution inconsistencies, and implementing automated and manual quality-control processes in Raiser's Edge NXT.

AI Pipeline Engineer **Mar 2025 — Aug 2025**
PowerKiosk, via UChicago Industry Capstone *Remote*

- Partnered with a digital energy marketplace to lead end-to-end system design for a 4-person team, delivering a solution that earned Best In Show for impact and execution quality across the full UChicago Applied Data Science capstone cohort.
- Transformed 1,000+ digital and scanned energy bills and contracts with no consistent schema into a fully queryable analytical layer via OCR, NLP information extraction, and vector embeddings ready for analyst-facing search and comparison.
- Replaced a manual document review process taking hours per query with a RAG-powered natural language interface returning results in seconds for energy market analysts, without sacrificing traceability back to source bill and contract documents.

Analytics Research Assistant **Sep 2023 — June 2024**
UCSB Baseball, University of California, Santa Barbara (NCAA D1) *Santa Barbara, CA*

- Built a predictive evaluation framework across 750+ players for a Division-1 program, isolating high-ROI recruiting targets from small-sample noise to inform coaching staff roster decisions during weekly recruiting discussions with the scouting staff.
- Synthesized video and radar tracking data into opponent scouting reports on high-leverage tendencies, becoming a repeatable input to game preparation for pitching and hitting matchup planning shared with coaching staff before each series.
- Modeled team bunting through Expected Value analysis to quantify run-scoring impact by game situation, turning coaching intuition into a data-backed framework that challenged prior in-game strategy assumptions across count and base-out states.

Projects

Independent Quantitative Research on MLB Pitch Tunneling **Jun 2025 — Present**
The University of Chicago *Remote*

- Led an end-to-end independent research project with no established feature framework, building one from first principles and testing competing spatiotemporal specs to achieve a 150%+ R^2 lift on swinging-strike prediction while preserving model.
- Applied causal inference techniques to identify the primary mechanism driving observed outcomes, demonstrating up to +23.4% lift in target behavioral responses within strategically defined regions of the pitch-tunnel feature space used in the study.
- Created interpretable visualizations translating complex statistical outputs from research experiments into clear, actionable insights for non-technical stakeholders while preparing findings for submission to a peer-reviewed baseball analytics journal.

Skills

Languages & Tools: Python, SQL, R, pandas, NumPy, DuckDB, Git, Jupyter, Statistical Analysis, Data Analysis
ML & Stats: Machine Learning, Scikit-learn, PyTorch, XGBoost, NLP, LLMs (RAG), Feature Engineering, Causal Inference, A/B Testing
Data & Deployment: ETL Pipelines, Data Cleaning, Data Visualization, Tableau, Streamlit, Matplotlib, FastAPI, Docker, AWS, GCP

Education

Master of Science, Applied Data Science, *The University of Chicago* **Aug 2024 — Aug 2025**
Bachelor of Science, Applied Mathematics, *University of California, Santa Barbara* **Sep 2020 — June 2024**